



Active Directory Home Lab

Design, Implementation and
Security Configuration of a
WindowsActive Directory
Environment

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For: Cybersecurity portfolio

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1. INTRODUCTION

Active Directory Domain Services (AD DS) is a Microsoft directory service used to centrally manage identities, computers, authentication and access within Windows-based environments.

This project involved designing and implementing a small enterprise-style Active Directory environment using Windows Server 2022, Windows 11 and Oracle VirtualBox. A Windows Server 2022 virtual machine was configured as the Domain Controller for the cyberlab.local domain, while a Windows 11 virtual machine was configured as a domain client.

The project focused not only on Active Directory deployment, but also on DNS configuration, organizational structure, user and group management, Group Policy, password security, account lockout controls, Windows Defender and security event monitoring.

The final stage of the project involved deliberately generating authentication failures and an account lockout to verify that Windows security auditing could detect and record these events.

2. PROJECT OBJECTIVES

The primary objectives of the project were to:

1. Deploy Windows Server 2022 in a virtualized environment.
2. Configure a Windows Server machine as an Active Directory Domain Controller.
3. Create and configure the cyberlab.local domain.
4. Configure DNS to support Active Directory name resolution.
5. Create Organizational Units for departmental and infrastructure management.
6. Create domain users and security groups.
7. Implement password and account lockout policies using Group Policy.
8. Configure Windows Defender security controls on the Windows 11 client.
9. Join the Windows 11 client to the Active Directory domain.
10. Configure and verify security auditing.
11. Generate and investigate failed authentication and account-lockout events.
12. Document troubleshooting processes and security findings.

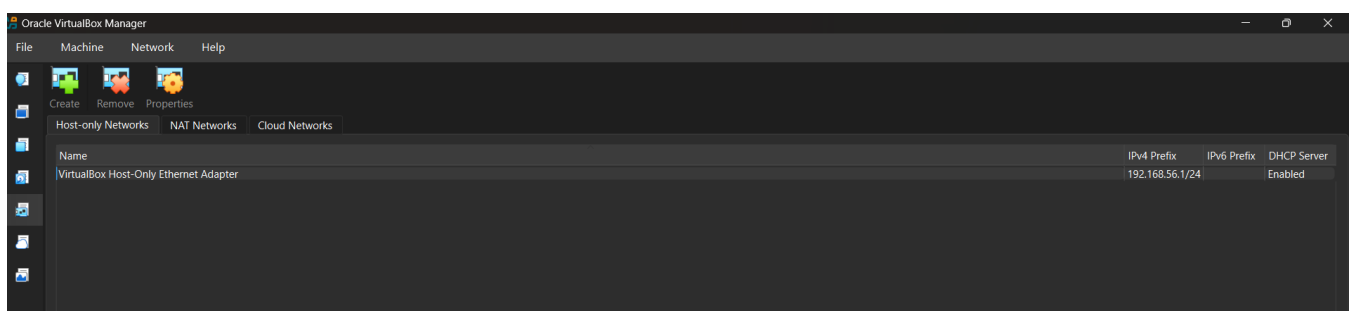
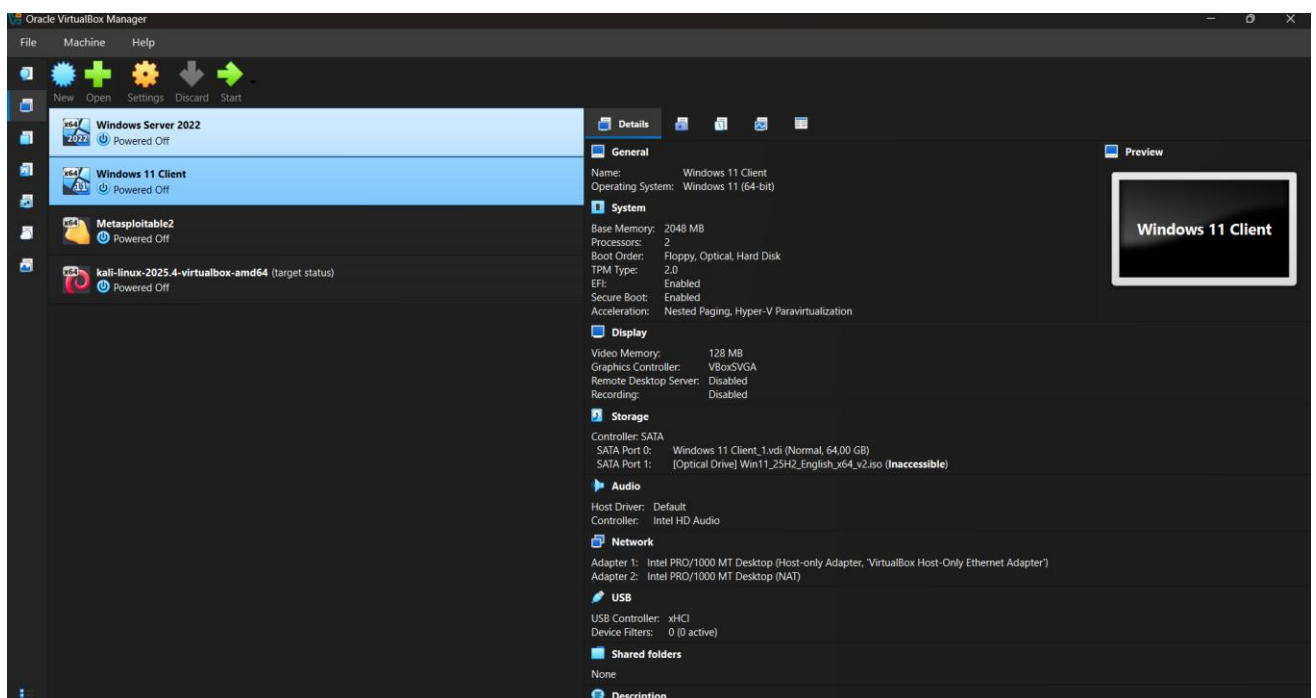
3. LAB ENVIRONMENT

3.1 Virtualization Platform

The environment was implemented using **Oracle VirtualBox**.

Virtualization allowed the Active Directory infrastructure to be deployed on a single physical computer while keeping the laboratory environment isolated from the normal host network.

The configured VirtualBox environment used a **Host-Only Adapter** for communication between the laboratory machines.



3.2 Virtual Machines

The main virtual machines used in the project were:

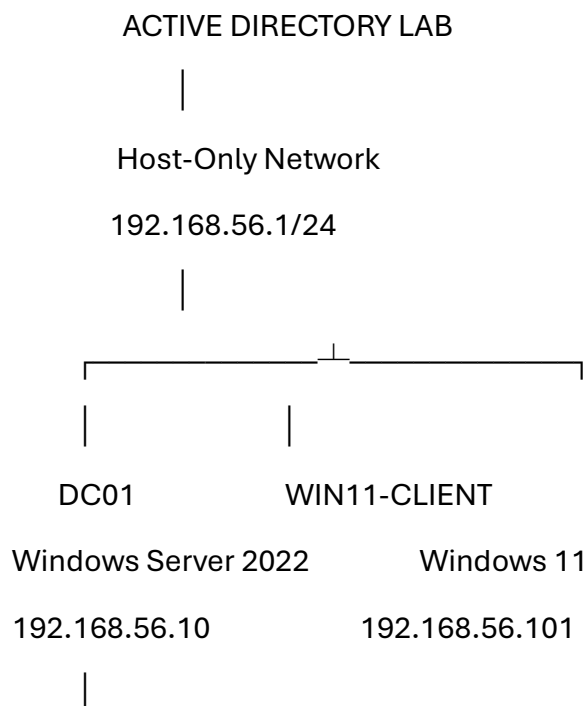
Machine	Operating System	Role	IPv4 Address
DC01	Windows Server	Domain controller/DNS server	192.168.56.10
Windows 11 Client	Windows 11	Domain client	192.168.56.101

3.3 Active Directory Configuration

Component	Configuration
Domain	cyberlab.local
Domain Controller	DC01
DC IPv4	192.168.56.10
Network	VirtualBox Host-Only
Client	Windows 11

3.4 Network Architecture

The Active Directory laboratory was implemented using VirtualBox Host-Only networking to provide an isolated network for communication between the Domain Controller and client systems. Windows Server 2022 was configured as the Domain Controller and DNS server with the IPv4 address 192.168.56.10. The Windows 11 client was connected to the same Host-Only network and configured with an IPv4 address in the 192.168.56.0/24 subnet. This configuration allowed the client to communicate with the Domain Controller while keeping the laboratory environment separated from the physical network.



AD DS + DNS

|

cyberlab.local

4. WINDOWS SERVER DEPLOYMENT

4.1 Windows Server Installation

A Windows Server 2022 virtual machine was created in Oracle VirtualBox.

The server formed the foundation of the laboratory because it was used as the Domain Controller. The operating system was installed and the server was configured with the hostname:

DC01

4.2 Server Naming

The name DC01 was selected to clearly identify the machine as the first Domain Controller in the environment.

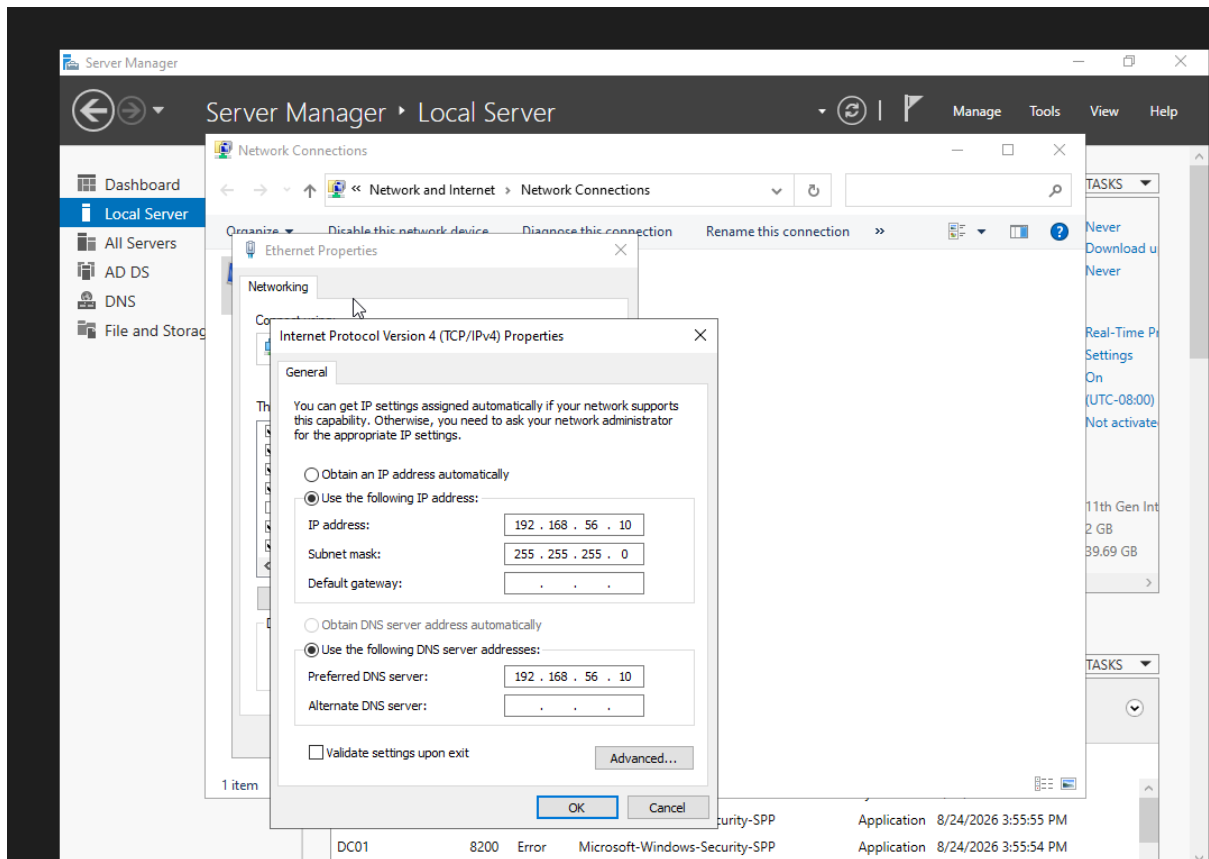
A consistent naming convention improves manageability and makes infrastructure easier to understand.

4.3 Static IP Configuration

The server was configured with a static IPv4 address:

192.168.56.10

The static address was important because infrastructure services such as Active Directory and DNS should have predictable network addresses.

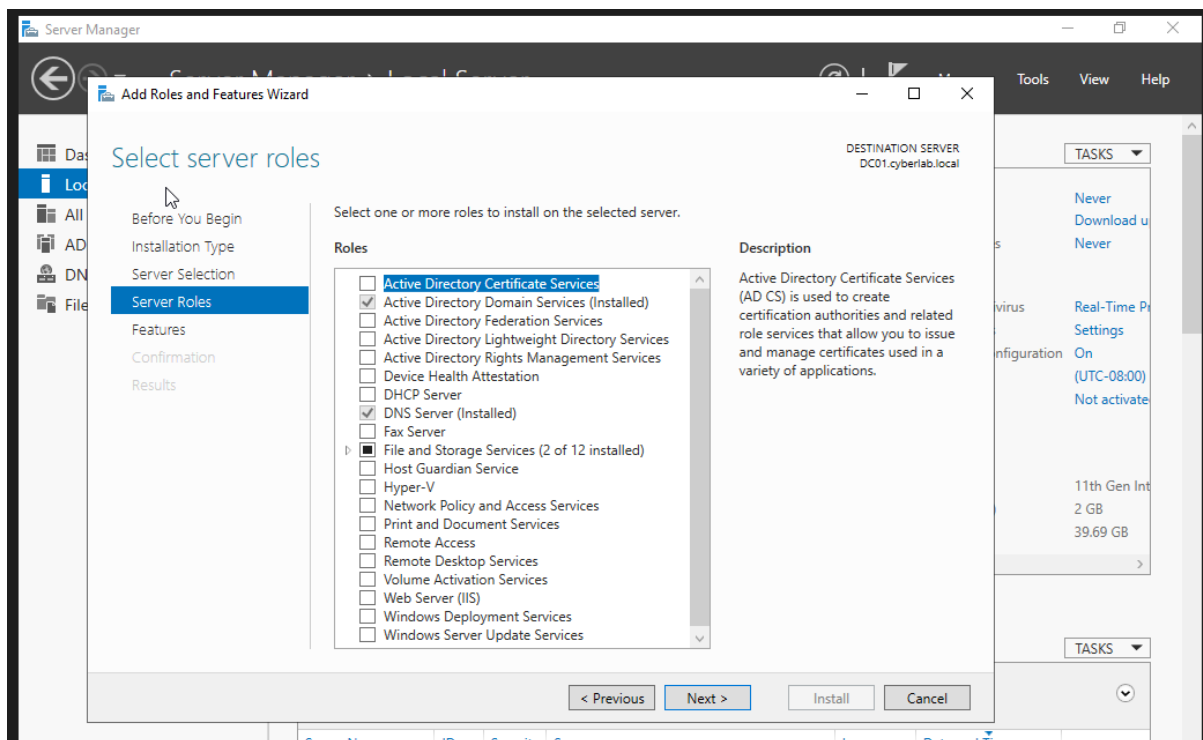


5. ACTIVE DIRECTORY DOMAIN SERVICES

5.1 AD DS Installation

The Active Directory Domain Services role was installed on Windows Server 2022.

This role provides the functionality required for centralized management of users, computers and domain resources.



5.2 Domain Controller Promotion

A new Active Directory forest was created using:

cyberlab.local

The resulting environment used:

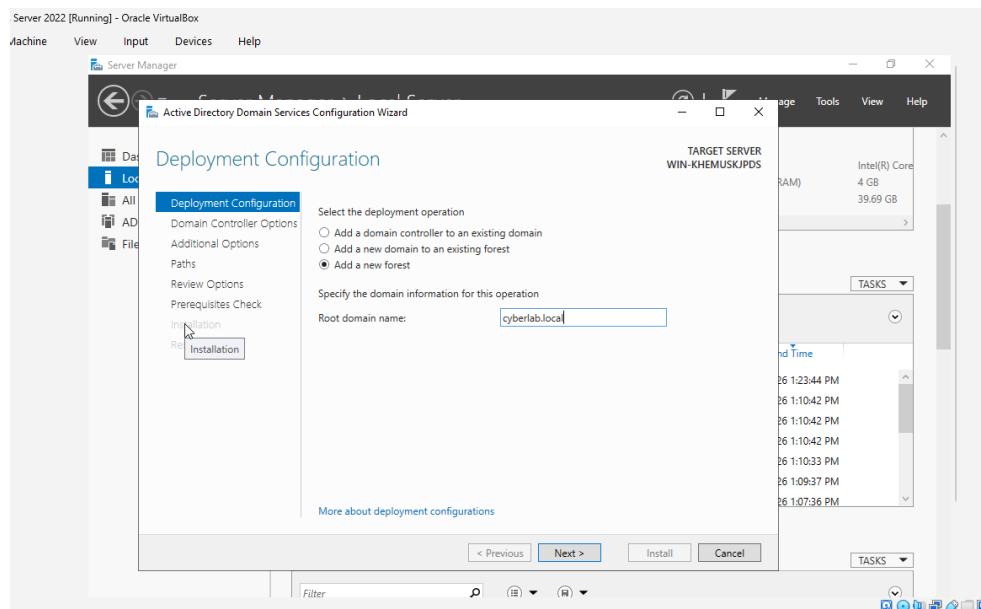
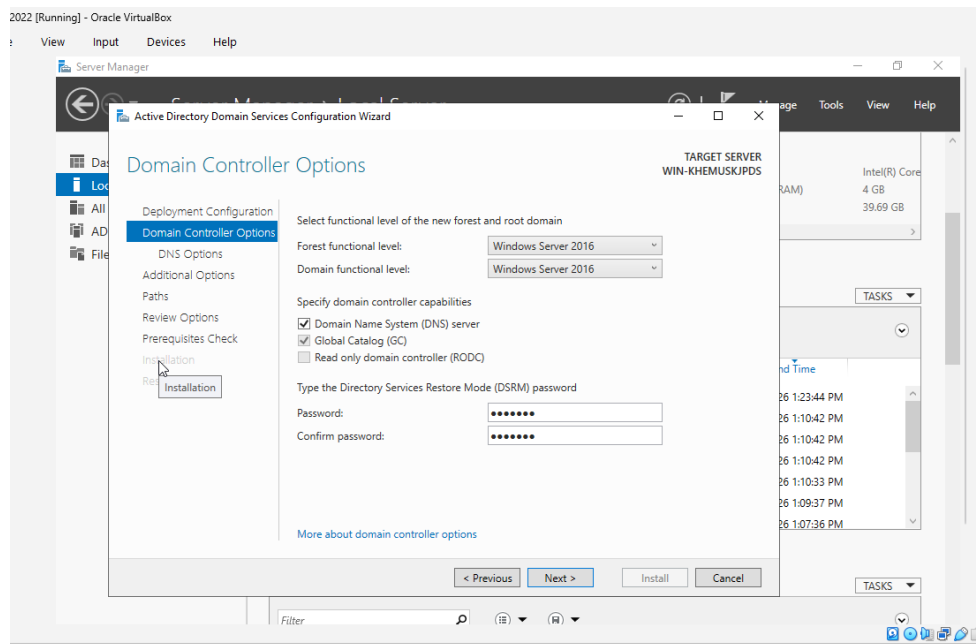
- Forest: cyberlab.local
- Domain: cyberlab.local
- Domain Controller: DC01

The server was promoted to a Domain Controller after the AD DS role had been installed.

The promotion process established:

- The Active Directory forest
- The Active Directory domain
- DNS integration
- Domain authentication services
- Directory services

The server was restarted after promotion to complete the configuration.

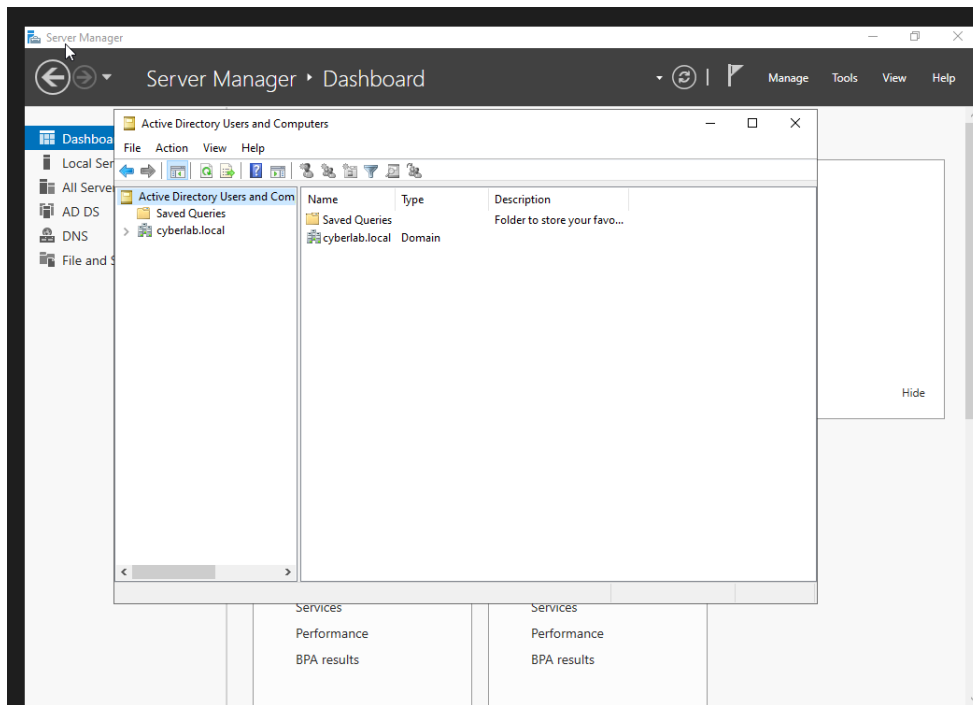


5.3 Domain Verification

After the server restarted, Active Directory Users and Computers was used to verify that the domain existed and was accessible.

The domain was displayed as:

cyberlab.local



6. DNS CONFIGURATION AND VERIFICATION

6.1 Importance of DNS

DNS is a critical dependency of Active Directory. Active Directory uses DNS to locate domain controllers and other domain services. If DNS is incorrectly configured, domain clients may experience problems locating the Domain Controller or joining and authenticating against the domain. For this reason, DNS was configured as part of the Domain Controller deployment.

6.2 DNS Configuration

The Domain Controller at:

192.168.56.10

provided DNS services for:

cyberlab.local

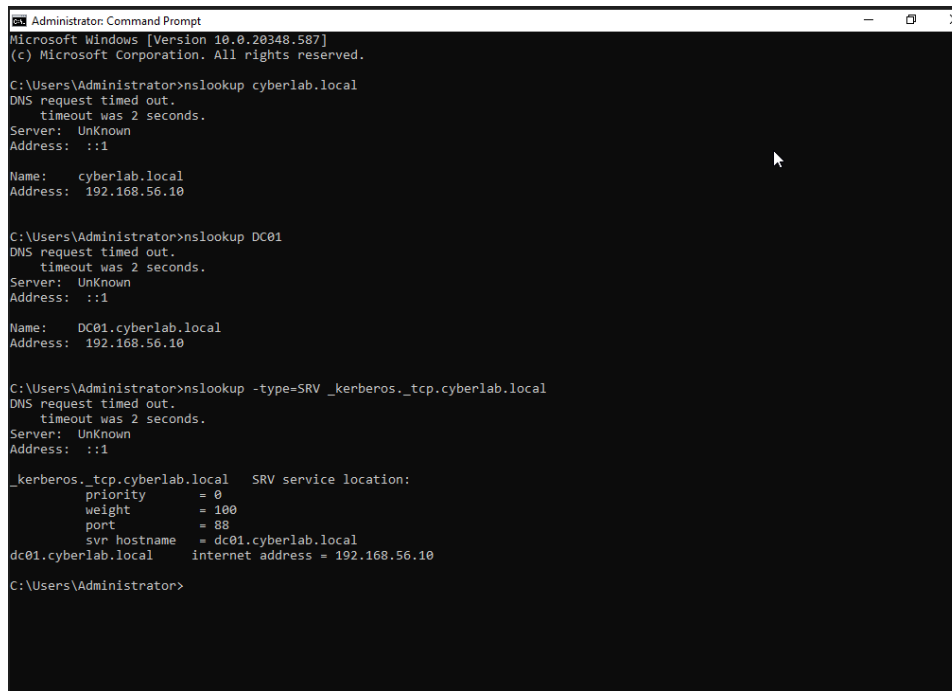
6.3 DNS Verification

DNS was tested with:

nslookup cyberlab.local

nslookup DC01

nslookup -type=SRV _kerberos._tcp.cyberlab.local



```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.20348.587]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>nslookup cyberlab.local
DNS request timed out.
    timeout was 2 seconds.
Server: UnKnown
Address: ::1

Name:   cyberlab.local
Address: 192.168.56.10

C:\Users\Administrator>nslookup DC01
DNS request timed out.
    timeout was 2 seconds.
Server: UnKnown
Address: ::1

Name:   DC01.cyberlab.local
Address: 192.168.56.10

C:\Users\Administrator>nslookup -type=SRV _kerberos._tcp.cyberlab.local
DNS request timed out.
    timeout was 2 seconds.
Server: UnKnown
Address: ::1

_kerberos._tcp.cyberlab.local    SRV service location:
        priority      = 0
        weight        = 100
        port          = 88
        svr hostname   = dc01.cyberlab.local
dc01.cyberlab.local             internet address = 192.168.56.10

C:\Users\Administrator>
```

6.4 Domain Controller Discovery

The following command was used:

nltest /dsgetdc:cyberlab.local

The successful resolution of the domain and discovery of the Domain Controller dc01.cyberlab.local confirmed that DNS was functioning correctly.

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>nltest /dsgetdc:cyberlab.local
    DC: \\DC01.cyberlab.local
    Address: \\192.168.56.10
    Dom Guid: fceacaa3-fbaf-434c-8c1d-2f417f047fc6
    Dom Name: cyberlab.local
    Forest Name: cyberlab.local
    Dc Site Name: Default-First-Site-Name
    Our Site Name: Default-First-Site-Name
    Flags: PDC GC DS LDAP KDC TIMESERV GTIMESERV WRITABLE DNS_DC DNS_DOMAIN DN
    S_FOREST CLOSE_SITE FULL_SECRET WS DS_8 DS_9 DS_10 KEYLIST
The command completed successfully
```

7. ACTIVE DIRECTORY STRUCTURE

7.1 Organizational Structure

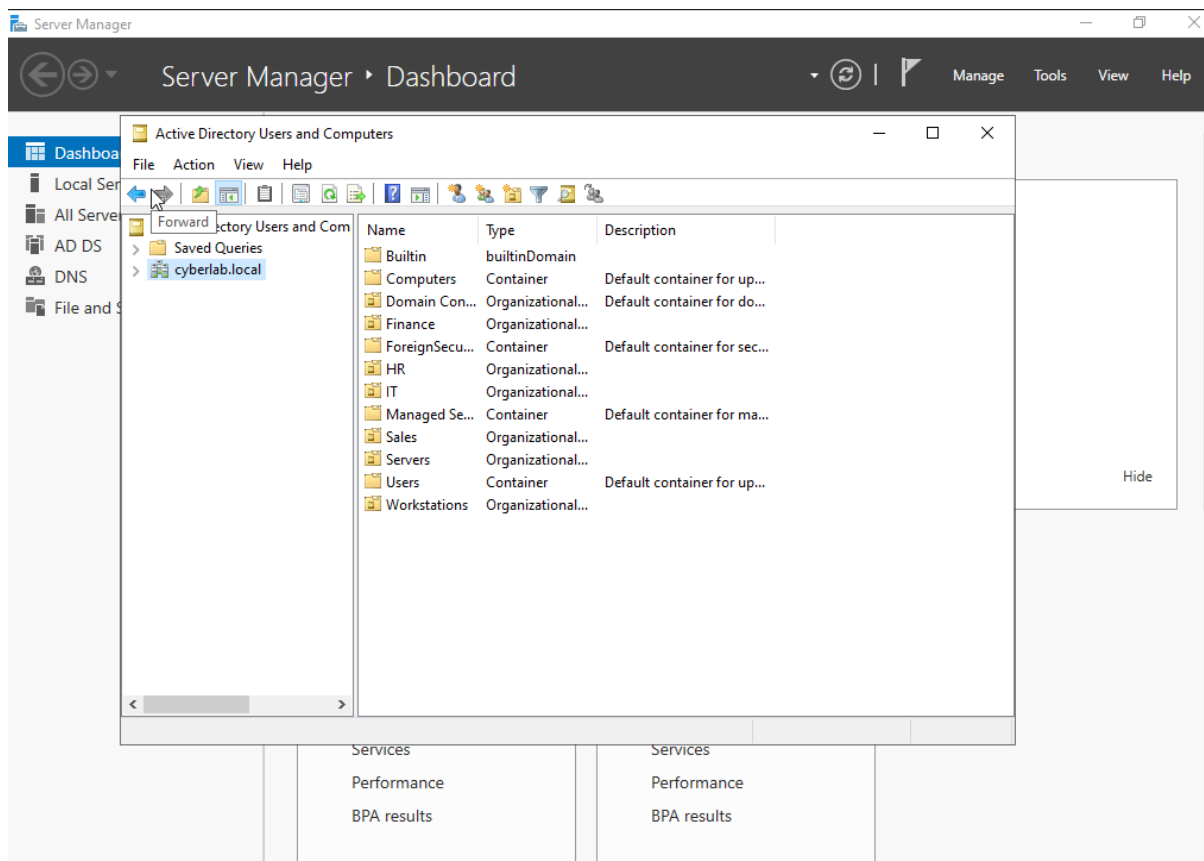
The following OUs were created:

OU	Purpose
IT	IT department
Finance	Finance department
HR	Human Resources
Sales	Sales department
Servers	Server computers
Workstations	Client computers

Organizational Units (OUs) provide a logical structure for organizing users and computers in Active Directory.

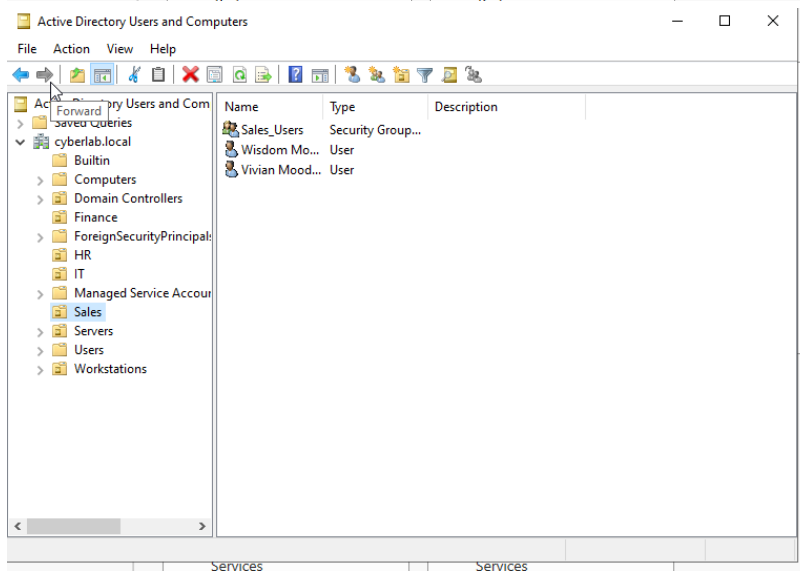
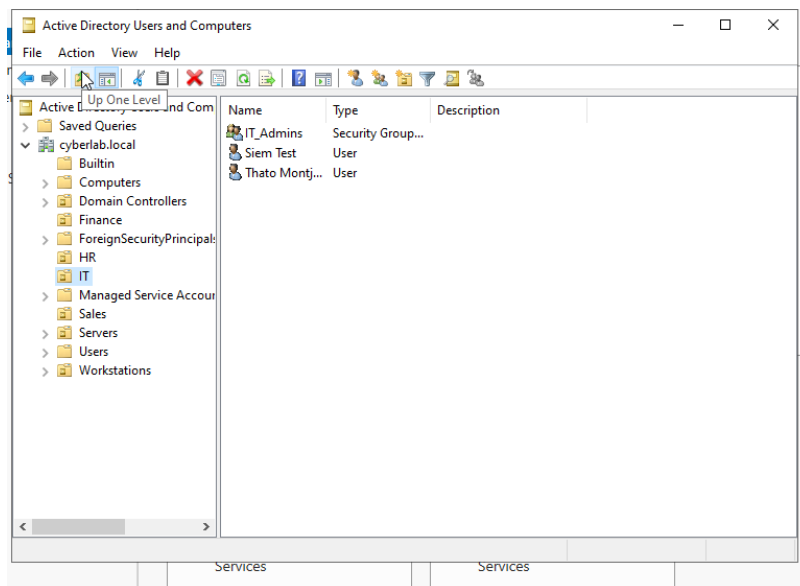
They are useful for:

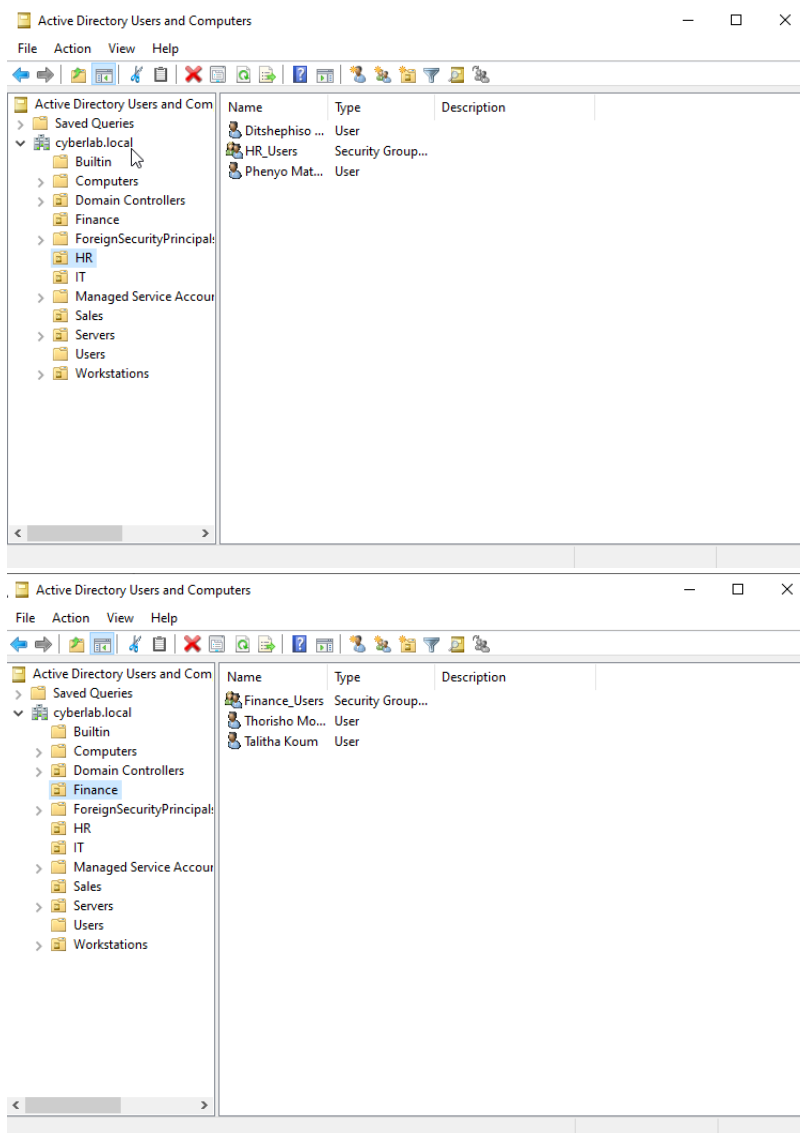
- Grouping users by department
- Organizing computers
- Applying Group Policy
- Simplifying administration
- Delegating administrative responsibilities



7.2 User Accounts

Domain user accounts were created through Active Directory Users and Computers. Accounts were placed into appropriate departmental OUs. This demonstrated centralized identity administration.





7.3 Security Groups

The security groups created during the lab were:

Security Group	Purpose
IT_Admins	IT administrative users
Sales_Users	Sales users
HR_Users	HR users
Finance_Users	Finance users

8. GROUP POLICY AND SECURITY HARDENING

8.1 Group Policy Configuration

Group Policy allows administrators to centrally manage configuration and security settings for domain users and computers. Instead of configuring each machine individually, administrators can apply policies through Active Directory.

8.2 Password Policy

Policy	Configuration
Minimum password age	1 day
Maximum password age	60 days
Minimum password length	12 characters
Password history	5 passwords

These settings were verified using:

net accounts

```
C:\Users\Administrator>net accounts
Force user logoff how long after time expires?:      Never
Minimum password age (days):                        1
Maximum password age (days):                        60
Minimum password length:                             12
Length of password history maintained:                5
```

8.3 Account Lockout Policy

Account lockout controls help reduce the effectiveness of repeated authentication attempts. The lab implemented an account lockout policy that would temporarily prevent authentication after a specified number of failed attempts.

```
Lockout threshold:                                   5
Lockout duration (minutes):                          15
Lockout observation window (minutes):                 15
Computer role:                                       WORKSTATION
The command completed successfully.
```

```
C:\Users\Administrator>
```

8.4 Group Policy Verification

The client policy was refreshed using:

`gpupdate /force`

The command completed successfully, confirming that the client could receive Group Policy from the domain.

```
C:\Users\Administrator>gpupdate /force
Updating policy...

Computer Policy update has completed successfully.
User Policy update has completed successfully.
```

9. WINDOWS 11 DOMAIN CLIENT

9.1 Windows 11 Configuration

A Windows 11 virtual machine was used as the client endpoint.

The machine was configured to communicate with the Domain Controller using the Host-Only network.

The client was configured to communicate with the Active Directory environment.

Connectivity was confirmed using network and DNS testing.

```
Administrator: Command Prompt
Host Name . . . . . : WIN11-CLIENT
Primary Dns Suffix . . . . . : cyberlab.local
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : cyberlab.local

Ethernet adapter Ethernet:

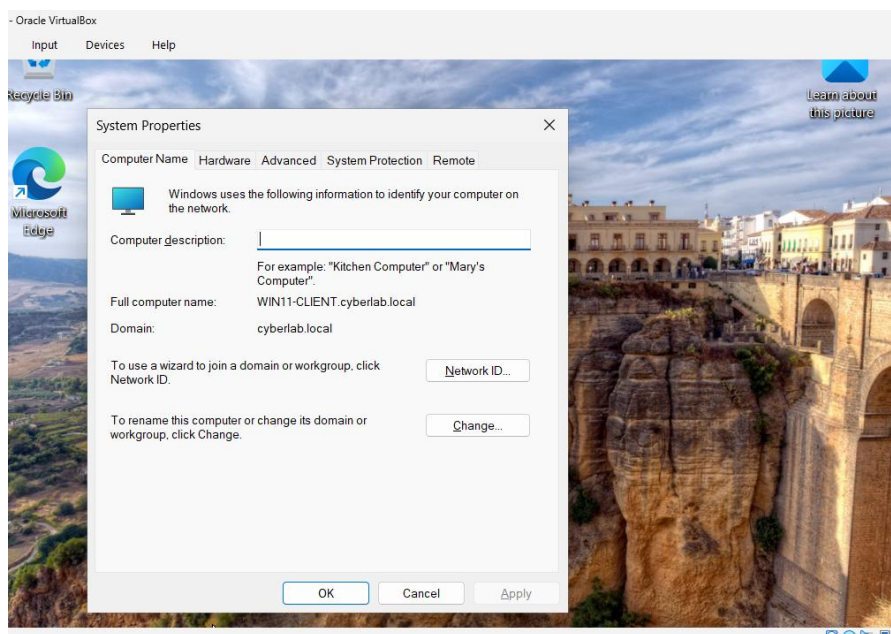
Connection-specific DNS Suffix . : 
Description . . . . . : Intel(R) PRO/1000 MT Desktop Adapter
Physical Address. . . . . : 08-00-27-DE-F3-32
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::88e4:2c71:9f94:3b5%12(Preferred)
IPv4 Address. . . . . : 192.168.56.101(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Tuesday, 25 August 2026 10:51:14
Lease Expires . . . . . : Tuesday, 25 August 2026 20:11:14
Default Gateway . . . . . : 
DHCP Server . . . . . : 192.168.56.100
DHCPv6 IAID . . . . . : 84410407
DHCPv6 Client DUID. . . . . : 00-01-01-00-32-0D-50-19-08-00-27-DE-F3-32
DNS Servers . . . . . : 192.168.56.10
NetBIOS over Tcpip. . . . . : Enabled
```

9.2 Domain Joining

The Windows 11 machine was joined to:

cyberlab.local

This allowed the client to authenticate against the Domain Controller and receive domain policies.



9.3 Domain Authentication

After joining the domain, domain authentication was tested using Active Directory user credentials.

Successful authentication demonstrated that:

- The client could communicate with the Domain Controller.
- DNS was functioning.
- Active Directory was operational.
- Domain authentication was working.

```
Administrator: Command Pro × + ▾  
Microsoft Windows [Version 10.0.26200.9168]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\Administrator>echo %LOGONSERVER%  
\\DC01  
  
C:\Users\Administrator>
```

9.4 Client Verification

```
Administrator: Command Prom...
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>ping 192.168.56.10

Pinging 192.168.56.10 with 32 bytes of data:
Reply from 192.168.56.10: bytes=32 time=3ms TTL=128
Reply from 192.168.56.10: bytes=32 time=3ms TTL=128
Reply from 192.168.56.10: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.56.10:
    Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 2ms
Control-C
^C
C:\Users\Administrator>nslookup cyberlab.local
DNS request timed out.
    timeout was 2 seconds.
Server:      Unknown
Address:     192.168.56.10

Name:       cyberlab.local
Address:    192.168.56.10

C:\Users\Administrator>

C:\Users\Administrator>gpresult /r

Microsoft (R) Windows (R) Operating System Group Policy Result tool v2.0
© Microsoft Corporation. All rights reserved.

Created on 2026/08/17 at 20:11:46

RSOP data for CYBERLAB\Administrator on WIN11-CLIENT : Logging Mode
-----

OS Configuration:      Member Workstation
OS Version: Start      10.0.26200

RSOP data for CYBERLAB\Administrator on WIN11-CLIENT : Logging Mode
-----

OS Configuration:      Member Workstation
OS Version:             10.0.26200
Site Name:              Default-First-Site-Name
Roaming Profile:         N/A
Local Profile:          C:\Users\Administrator
Connected over a slow link?: No

COMPUTER SETTINGS
-----
CN=WIN11-CLIENT,OU=Workstations,DC=cyberlab,DC=local
Last time Group Policy was applied: 2026/08/17 at 20:11:10
Group Policy was applied from: DC01.cyberlab.local
Group Policy slow link threshold: 500 kbps
Domain Name:             CYBERLAB
Domain Type:             Windows 2008 or later

Applied Group Policy Objects
-----
CYBERLAB - Workstation Security Baseline
Default Domain Policy
```

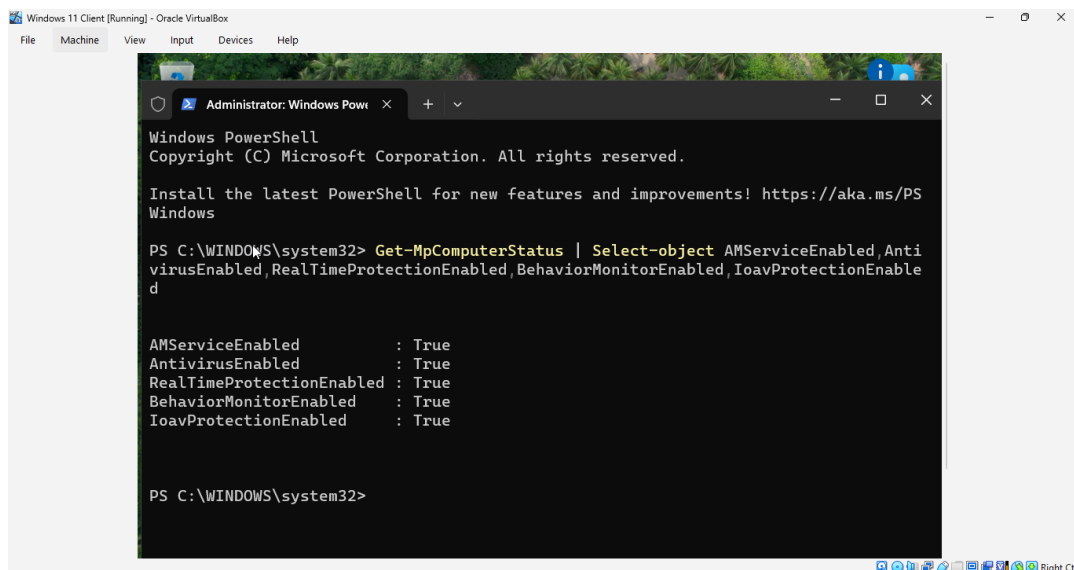
10. WINDOWS DEFENDER SECURITY CONFIGURATION

10.1 Defender Configuration

Windows Defender was configured on the Windows 11 client as part of the endpoint security controls implemented in the Active Directory laboratory. The configuration was intended to provide protection against malware and other potentially unwanted threats. Security settings were reviewed and verified on the Windows 11 client to ensure that the required protection features were enabled.

10.2 Protection Verification

The configuration was checked using PowerShell to confirm that the relevant Defender protection settings were enabled.



The screenshot shows a Windows 11 client running in Oracle VM VirtualBox. An Administrator PowerShell window is open, displaying the output of the command `Get-MpComputerStatus | Select-object AMServiceEnabled, AntivirusEnabled, RealTimeProtectionEnabled, BehaviorMonitorEnabled, IoavProtectionEnabled`. The output shows that all five protection features are enabled (True).

```
Windows 11 Client [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> Get-MpComputerStatus | Select-object AMServiceEnabled, AntivirusEnabled, RealTimeProtectionEnabled, BehaviorMonitorEnabled, IoavProtectionEnabled

AMServiceEnabled      : True
AntivirusEnabled       : True
RealTimeProtectionEnabled : True
BehaviorMonitorEnabled : True
IoavProtectionEnabled  : True

PS C:\WINDOWS\system32>
```

11. SECURITY AUDITING AND EVENT MONITORING

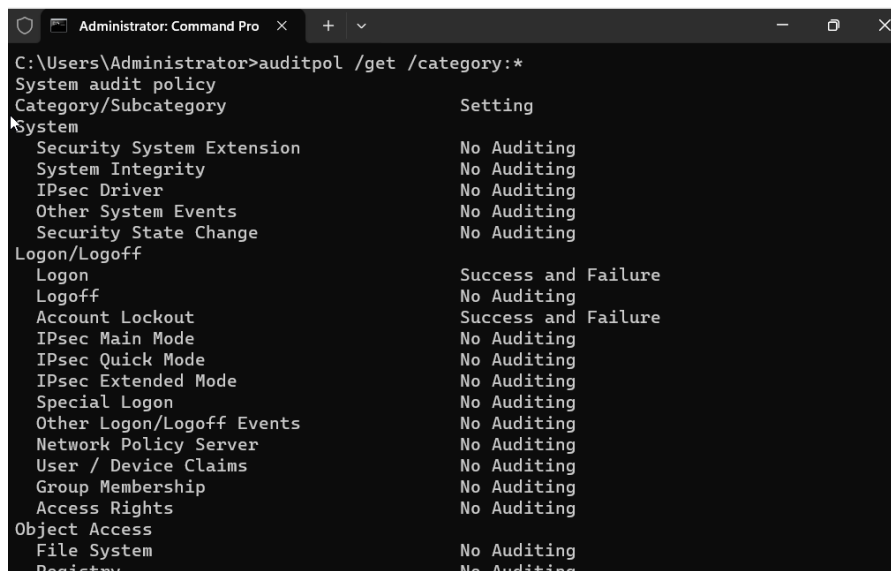
11.1 Audit Policy

Security auditing was configured to record successful and failed security events.

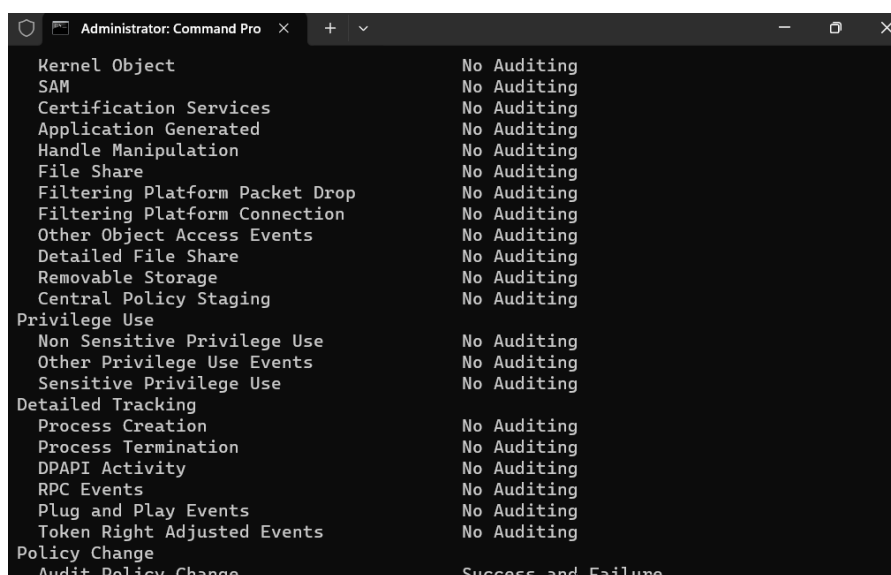
Verification was performed using:

```
auditpol /get /category:*
```

The relevant audit policies were configured for Success and Failure.



```
C:\Users\Administrator>auditpol /get /category:*
System audit policy
Category/Subcategory          Setting
System
  Security System Extension    No Auditing
  System Integrity             No Auditing
  IPsec Driver                 No Auditing
  Other System Events          No Auditing
  Security State Change        No Auditing
Logon/Logoff
  Logon                        Success and Failure
  Logoff                       No Auditing
  Account Lockout              Success and Failure
  IPsec Main Mode              No Auditing
  IPsec Quick Mode             No Auditing
  IPsec Extended Mode          No Auditing
  Special Logon                No Auditing
  Other Logon/Logoff Events     No Auditing
  Network Policy Server        No Auditing
  User / Device Claims         No Auditing
  Group Membership             No Auditing
  Access Rights                No Auditing
Object Access
  File System                  No Auditing
  Registry                     No Auditing
```



```
Kernel Object                  No Auditing
SAM                            No Auditing
Certification Services         No Auditing
Application Generated          No Auditing
Handle Manipulation            No Auditing
File Share                     No Auditing
Filtering Platform Packet Drop No Auditing
Filtering Platform Connection  No Auditing
Other Object Access Events     No Auditing
Detailed File Share            No Auditing
Removable Storage              No Auditing
Central Policy Staging         No Auditing
Privilege Use
  Non Sensitive Privilege Use   No Auditing
  Other Privilege Use Events    No Auditing
  Sensitive Privilege Use       No Auditing
Detailed Tracking
  Process Creation              No Auditing
  Process Termination           No Auditing
  DPAPI Activity                No Auditing
  RPC Events                    No Auditing
  Plug and Play Events          No Auditing
  Token Right Adjusted Events   No Auditing
Policy Change
  Audit Policy Change           Success and Failure
```

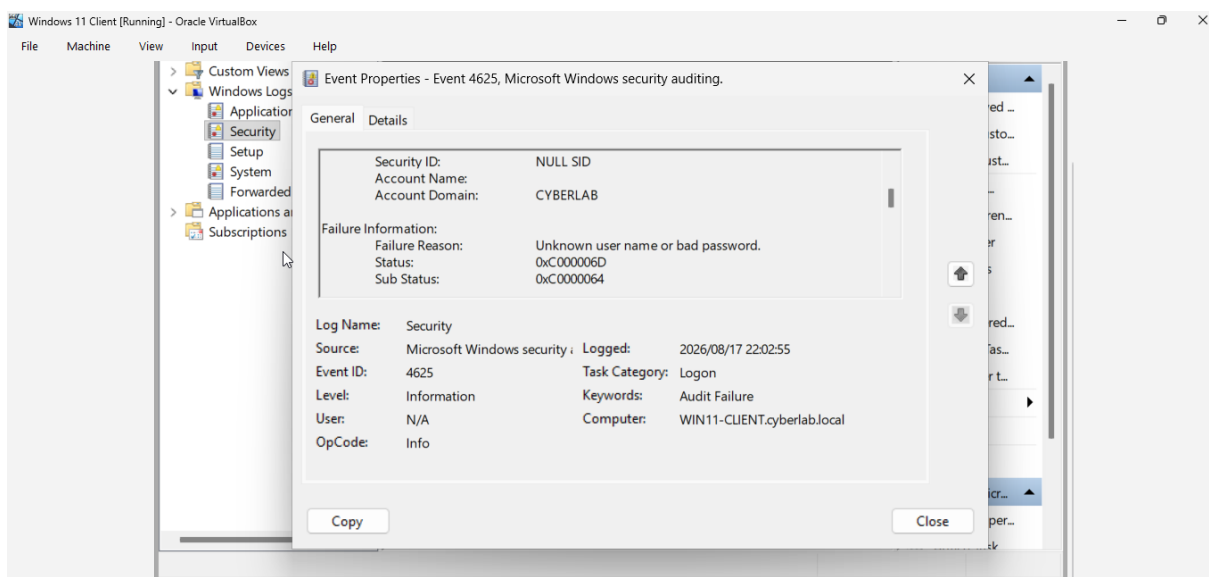
Token Right Adjusted Events	No Auditing
Policy Change	
Audit Policy Change	Success and Failure
Authentication Policy Change	Success and Failure
Authorization Policy Change	No Auditing
MPSSVC Rule-Level Policy Change	No Auditing
Filtering Platform Policy Change	No Auditing
Other Policy Change Events	No Auditing
Account Management	
Computer Account Management	No Auditing
Security Group Management	Success and Failure
Distribution Group Management	No Auditing
Application Group Management	No Auditing
Other Account Management Events	No Auditing
User Account Management	Success and Failure
DS Access	
Directory Service Access	No Auditing
Directory Service Changes	No Auditing
Directory Service Replication	No Auditing
Detailed Directory Service Replication	No Auditing
Account Logon	
Kerberos Service Ticket Operations	No Auditing
Other Account Logon Events	No Auditing
Kerberos Authentication Service	No Auditing
Credential Validation	Success and Failure

12. SECURITY EVENT INVESTIGATION

12.1 Failed Authentication – Event ID 4625

A controlled failed authentication test was performed. Windows recorded multiple failed logon events.

The investigation identified:



These events demonstrated that Windows was recording authentication failures.

12.2 Account Lockout – Event ID 4740

A dedicated test account:

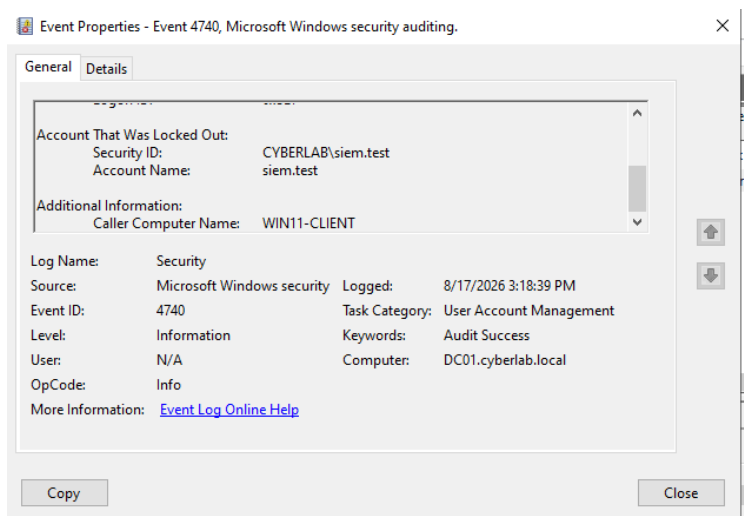
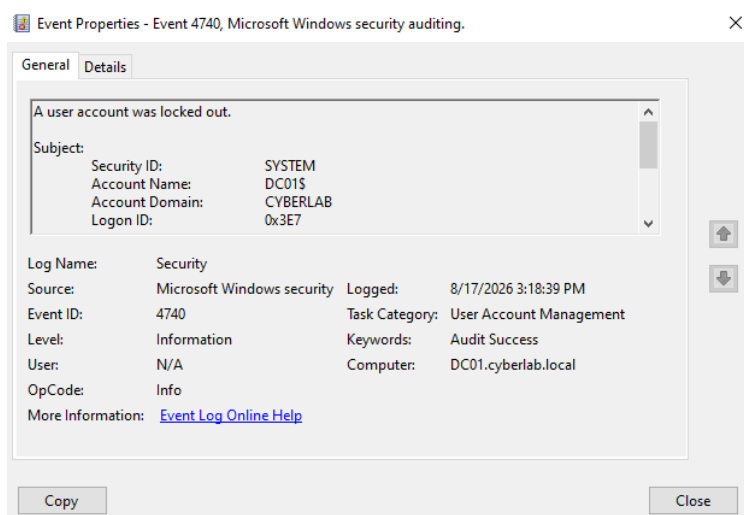
CYBERLAB\siem.test

was created for the lockout test.

Five unsuccessful authentication attempts were deliberately generated, resulting in the account being locked according to the configured policy.

The screenshot shows the 'Siem Test Properties' dialog box with the 'General' tab selected. The 'First name' field contains 'Siem' and the 'Last name' field contains 'Test'. The 'Display name' field shows 'Siem Test'. The 'Description' and 'Office' fields are empty. The 'Telephone number' and 'Web page' fields are also empty, each with an 'Other...' button. The 'E-mail' field is empty. At the bottom are 'OK', 'Cancel', 'Apply', and 'Help' buttons.

The screenshot shows the 'Siem Test Properties' dialog box with the 'Account' tab selected. The 'User logon name' field contains 'siem.test' and the domain dropdown shows '@cyberlab.local'. The 'User logon name (pre-Windows 2000)' field contains 'CYBERLAB\'. Below these are 'Logon Hours...' and 'Log On To...' buttons. A checkbox labeled 'Unlock account. This account is currently locked out on this Active Directory Domain Controller.' is checked. Under 'Account options', there are four unchecked checkboxes: 'User must change password at next logon', 'User cannot change password', 'Password never expires', and 'Store password using reversible encryption'. Under 'Account expires', the 'Never' radio button is selected. The 'End of:' option is also visible with a date field showing 'Thursday, September 24, 2026'. At the bottom are 'OK', 'Cancel', 'Apply', and 'Help' buttons.



12.3 Findings

The security-event investigation demonstrated that the Active Directory environment was capable of detecting both individual authentication failures and account-lockout conditions. Event ID 4625 provided evidence of failed authentication, while Event ID 4740 identified the account that had been locked and the workstation from which the lockout originated. These results confirmed that the configured security policies and auditing mechanisms were functioning as intended.

13. TROUBLESHOOTING AND CHALLENGES

Challenge	Investigation	Resolution
DNS resolution problems	nslookup, IP configuration	Corrected DNS configuration
ERROR_NO_SUCH_DOMAIN	nltest, DNS and connectivity testing	Corrected domain/DNS configuration
Windows client domain-join issue	Checked Windows edition	Corrected Windows client configuration
Event Viewer access denied	Checked user privileges	Opened Event Viewer with appropriate administrative privileges

Troubleshooting followed a layered approach, beginning with physical and virtual network connectivity, followed by IP configuration, DNS resolution, Domain Controller discovery, Active Directory services, Group Policy and authentication. This approach helped isolate configuration problems without unnecessarily changing functioning components.

14. SKILLS AND TECHNOLOGIES USED

Technical Skills:

- Windows Server administration
- Active Directory
- DNS
- Group Policy

- User and group management
- Identity and access management
- Windows security hardening
- Network troubleshooting
- Security auditing
- Event log analysis

Technologies:

Technology	Purpose
Windows Server 2022	Domain Controller
Windows 11	Domain client
Active Directory	Identity and access management
DNS	Name resolution and AD service discovery
Group Policy	Centralized security configuration
VirtualBox	Virtualization
Event Viewer	Security event investigation
Windows Defender	Endpoint protection
PowerShell/Command Prompt	Configuration and verification

15. LESSONS LEARNED

15.1 DNS is fundamental to Active Directory

Active Directory relies on DNS for domain-controller and service discovery.

15.2 Security policies must be tested

Configuring a password or lockout policy does not prove that it works. The `siem.test` exercise demonstrated the importance of testing controls.

15.3 Logs provide valuable security evidence

Events such as 4625 and 4740 can provide information about authentication failures and account lockouts.

15.4 Troubleshooting requires a layered approach

Network connectivity, IP addressing, DNS, Domain Controller discovery and authentication should be checked systematically.

15.5 Practical labs connect theory to real systems

The project demonstrated how enterprise identity, networking and endpoint-security concepts interact in a functioning environment.

16. CONCLUSION

This project successfully demonstrated the design and implementation of a small enterprise-style Active Directory environment using Windows Server 2022, Windows 11 and Oracle VirtualBox. The Windows Server machine was configured as the Domain Controller for the cyberlab.local domain, with DNS configured to support Active Directory service discovery and domain authentication.

The environment was further developed through the creation of Organizational Units, user accounts and security groups, followed by the implementation of password, account-lockout and endpoint-security controls. Group Policy was used to centrally enforce security requirements, and the Windows 11 workstation was successfully integrated into the domain.

The security-monitoring component demonstrated the practical value of Windows auditing through the investigation of Event ID 4625 and Event ID 4740. These tests confirmed that failed authentication attempts and account-lockout activity could be detected and investigated using Windows Security logs.

Overall, the project provided practical experience in Windows infrastructure administration, Active Directory, DNS, Group Policy, identity and access management, endpoint security, troubleshooting and security-event analysis. It also established a foundation for future work involving centralized SIEM and log analysis.